Daily Number Theory Drill #4

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Prime Factor Algebra · Difference of Squares · Bounding Factors.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Prime factors, divisibility rules, and quick remainders.

1. Find the largest prime factor of 1001.

2. What is the last digit of 2^{2026}?

3. What is the remainder when $2^{10} + 3^{10}$ is divided by 5?

4. Find the greatest common divisor of 120 and 84.

5. How many positive integer factors does 144 have?

6. What is the remainder when $11 \times 12 \times 13$ is divided by 10?

7. What is the smallest positive integer divisible by $2, 3, 4, 5$, and 6?

8. Find the sum of the prime factors of 210.

9. What is the remainder when $10^{2026} - 1$ is divided by 9?

10. Find the value of k if 2^k is the highest power of 2 dividing 20!.

Section B Manipulation Drills (≤50 s each)

Modulo arithmetic properties, CRT basics, and divisor counting.

1. Find the smallest positive integer n such that $15n$ is a perfect square.

2. Find all single digits d such that the four-digit number $5d72$ is divisible by 9.

3. Find the number of trailing zeros in 50!.

4. What is the remainder when 5^{2025} is divided by 7?

5. If $x \equiv 3 \pmod 7$, find the remainder when $x^2 + 4x + 5$ is divided by 7.

6. Find the smallest positive integer that leaves a remainder of 2 when divided by 3, and 3 when divided by 5.

7. Find the sum of all positive integer divisors of 100.

8. What is the last digit of $3^{101} + 7^{101}$?

9. Find the number of positive divisors of $2^3 \times 3^4 \times 5$.

10. Find the largest three-digit number that is a multiple of 11 and 13.

Section C Substitution & Structure (≤90 s each)

Look for algebraic structures like difference of squares and combinatorial counting strategies.

1. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 45$. *(after SMC 2018 Q15)*

2. The prime number p can be expressed as $p = a^2 - b^2$, where a and b are also prime numbers. Find the value of p. *(after SMC 2011 Q19)*

3. Find the number of integer pairs (x, y) satisfying $x^2 - y^2 = 2018$.

4. How many ordered pairs of positive integers (x, y) satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$? *(after SMC 2015 Q21)*

5. Find all integer values of x for which $\frac{x^2+5x+14}{x+3}$ is an integer. *(after SMC 2019 Q20)*

6. How many divisors of 10^{10} are perfect squares? *(after SMC 2021 Q22)*

7. How many positive integers $n \leq 1000$ are not divisible by $2, 3$, or 5? *(after SMC 2017 Q19)*

8. Given that p and q are prime numbers satisfying $p = q^2 - 36$, find the value of p. *(after SMC 2014 Q22)*

Section D Challenge (BMO1 difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find all pairs of positive integers (m, n) such that $3^m - 2^n = 1$. *(after BMO1 2008 Q1)*

2. For how many positive integers n is $n^4 + 4^n$ a prime number? *(after BMO1 2013 Q2)*

3. A sequence is defined by $a_1 = 1$, $a_2 = 1$, and $a_{n+2} = a_{n+1} + a_n$. What is the remainder when a_{2026} is divided by 3? *(after SMC 2022 Q25)*

4. Let p, q, and r be prime numbers such that $p = 9q^2 - r^2$. Find the value of p. *(after BMO1 2010 Q1)*

5. Find the largest positive integer n such that 2^n divides $3^{1024} - 1$. *(after BMO1 2019 Q3)*

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths